Practical Work: Out-of-Distribution Detection, OOD Scoring Methods, and Neural Collapse

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1 Introduction

In this practical work, we train a ResNet-18 classifier on CIFAR-100 and explore two closely related topics: Out-of-Distribution (OOD) detection and the Neural Collapse phenomenon. On the OOD side, we implement and compare several scoring methods that try to distinguish in-distribution samples from unseen ones. On the Neural Collapse side, we study the geometric structure that spontaneously emerges in the network's last layer once it has been trained to near-zero loss, and we show how this structure can be exploited for OOD detection through the NECO method.

2 Training a ResNet-18 on CIFAR-100

We use a standard ResNet-18 adapted for 32×32 images: we replace the initial 7×7 convolution with a 3×3 one and remove the max pooling layer that would otherwise downsample the already small inputs too aggressively. The model has about 11.2M parameters and is trained for 100 epochs with SGD (momentum 0.9, weight decay 5×10^{-4}) and a cosine annealing schedule starting from a learning rate of 0.1.

Metric	**Value**
Final Train Accuracy	99.96%
Final Test Accuracy	77.13%
Final Train Loss	0.0106
Final Test Loss	0.9014
Parameters	11,220,132

Table 1: Training summary.

As shown in Figure 1, the training loss drops to near zero while the test loss stabilizes around 0.9, and the training accuracy reaches almost 100%. This gap between train and test performance is typical for CIFAR-100 and, importantly, it means the model has entered the *terminal phase of training* — the regime where Neural Collapse is expected to appear.

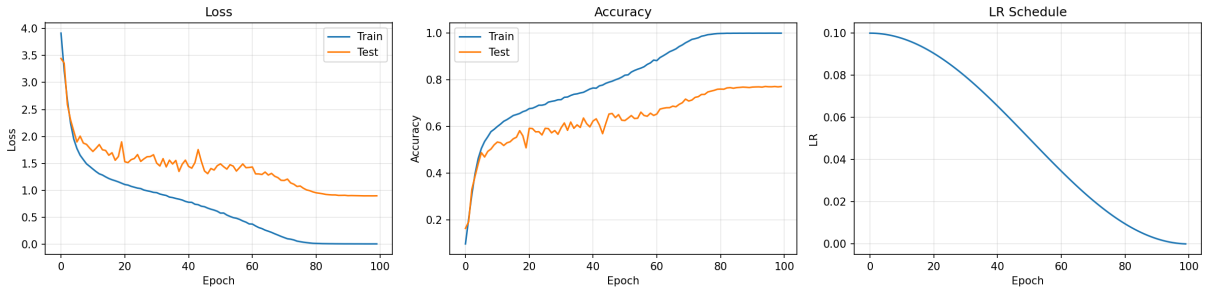


Figure 1: Training curves over 100 epochs. Left: loss (train vs. test). Center: accuracy (train vs. test). Right: cosine annealing learning rate schedule.

3 OOD Detection Methods

We use CIFAR-100 as the in-distribution (ID) dataset and SVHN as the out-of-distribution (OOD) dataset. For each method, we compute scores on both datasets and evaluate using AUROC (higher is better) and FPR@95 — the false positive rate when the true positive rate is 95% (lower is better). In all methods below, we adopt the convention that a **higher score indicates a sample is more likely ID**.

3.1 Maximum Softmax Probability (MSP)

The simplest baseline for OOD detection uses the maximum softmax probability as a confidence score:

$$s_{\text{MSP}}(x) = \max_c \ \text{softmax}(z_c(x)) = \max_c \frac{e^{z_c(x)}}{\sum_j e^{z_j(x)}}$$

The intuition is that a well-calibrated classifier should assign lower maximum probability to inputs it has not been trained on. However, neural networks are known to produce overconfident predictions even on OOD data, which limits this method. Indeed, the MSP score distribution plot (Figure 2, top-left) shows that both ID and OOD samples receive very high softmax values, with most scores concentrated near 1.0, confirming that the model is overconfident on OOD data.

Results. MSP achieves an AUROC of **0.8508** and FPR@95 of **0.6606**.

3.2 Maximum Logit Score

Instead of applying softmax, we directly use the maximum logit (pre-softmax) value:

$$s_{\text{MaxLogit}}(x) = \max_c \ z_c(x)$$

This avoids the saturation effect of softmax, where differences in logit magnitudes are compressed. The raw logit retains more information about the model's confidence. In the score distributions (Figure 2, top-centre), we observe that ID logits tend to be shifted to the right compared to OOD, though there is still significant overlap.

Results. Max Logit improves over MSP with an AUROC of **0.8807** and FPR@95 of **0.6292**, confirming that working with logits directly provides a better signal for OOD detection.

3.3 Energy Score

The energy score summarises the total evidence across all classes using the LogSumExp function:

$$E(x) = -\log \sum_c e^{z_c(x)} = -\text{LSE}(z(x))$$

Lower energy (more negative) indicates stronger total evidence, meaning the sample is more likely ID. We negate this quantity so that higher values correspond to ID. The energy can also be tempered with a temperature parameter T: $E_T(x) = -T \log \sum_c e^{z_c(x)/T}$.

Unlike MaxLogit which only considers the dominant class, the energy score aggregates evidence across all classes, providing a more holistic measure of model confidence.

Results. Energy achieves an AUROC of **0.8846** and FPR@95 of **0.6041**, slightly outperforming Max Logit. The improvement comes from capturing the full logit distribution rather than just the maximum.

3.4 Mahalanobis Distance Score

This method works in the penultimate feature space rather than the output space. For each class c, we compute the mean feature vector μ_c from training data and estimate a shared covariance matrix Σ. The Mahalanobis distance to the nearest class is:

$$d_{\text{Maha}}(x) = \min_c \, (f(x) - \mu_c)^\top \Sigma^{-1} (f(x) - \mu_c)$$

Larger distance means the sample's features are far from all known class distributions, indicating OOD. We negate the distance so that higher scores correspond to ID.

Results. Mahalanobis achieves an AUROC of only **0.7847** and FPR@95 of **0.8291**, performing worse than the logit-based methods. The score distribution (Figure 2, top-right) shows strongly overlapping distributions, explaining the poor performance. This is likely due to the Gaussian assumption not holding well in the 512-dimensional feature space, and the shared covariance being a poor approximation for some classes. With only 500 samples per class to estimate a 512×512 covariance, the estimation is inherently noisy.

3.5 ViM (Virtual-logit Matching)

ViM combines logit-based and feature-based information. The key idea is to decompose test features into a component that lies within the principal subspace of ID training features and a residual component:

1. Compute the principal subspace P via PCA on the centred ID training features (top d eigenvectors).
2. The **residual** measures how much of the feature falls outside this subspace: $r(x) = \|f(x) - PP^\top f(x)\|$.
3. Combine with the energy score: $\text{ViM}(x) = \text{Energy}(x) - \alpha \cdot r(x)$.

OOD samples tend to have larger residuals because they do not lie in the learned ID feature subspace, making this a powerful hybrid approach.

Results. ViM achieves the **best overall performance** with an AUROC of **0.9027** and FPR@95 of **0.5163**. The score distribution (Figure 2, bottom-left) shows the cleanest separation between ID and OOD of all methods. By combining output-level evidence (energy) with feature-level geometry (residual), it outperforms all purely logit-based or purely feature-based methods.

3.6 Comparison of OOD Methods

Table 2 summarises all OOD detection results. Among purely output-based methods, the progression MSP $\rightarrow$ Max Logit $\rightarrow$ Energy shows consistent improvement as we move from softmax probabilities to raw logits to the full LogSumExp aggregation. The Mahalanobis distance underperforms here, likely due to the difficulty of estimating a reliable covariance matrix in high

dimensions with limited per-class data. ViM achieves the best results by combining both information sources, confirming that hybrid methods that leverage both logit-level and feature-level signals are most effective.

Method	**Type**	**AUROC ↑**	**FPR@95 ↓**
MSP	Output-based	0.8508	0.6606
Max Logit	Output-based	0.8807	0.6292
Energy	Output-based	0.8846	0.6041
Mahalanobis	Feature-based	0.7847	0.8291
ViM	Hybrid	**0.9027**	**0.5163**
NECO (raw)	Feature-based	0.8477	0.5813
NECO (rescaled)	Hybrid	0.8834	0.6208

Table 2: Comparison of OOD detection methods (CIFAR-100 vs SVHN).

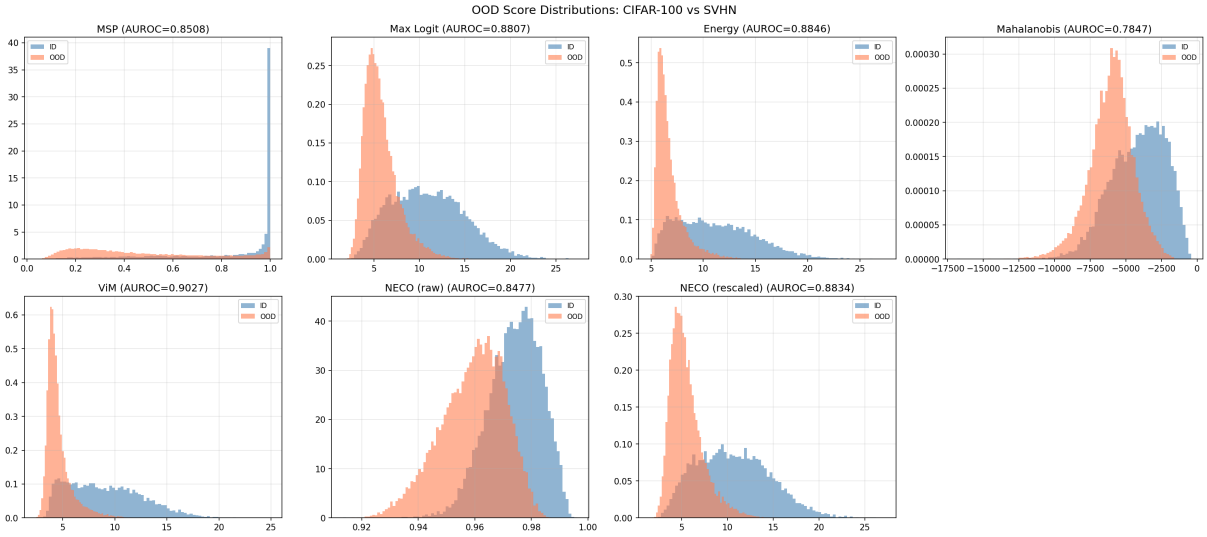


Figure 2: Score distributions for ID (CIFAR-100, blue) and OOD (SVHN, red) for each method. Better methods show more separated distributions.

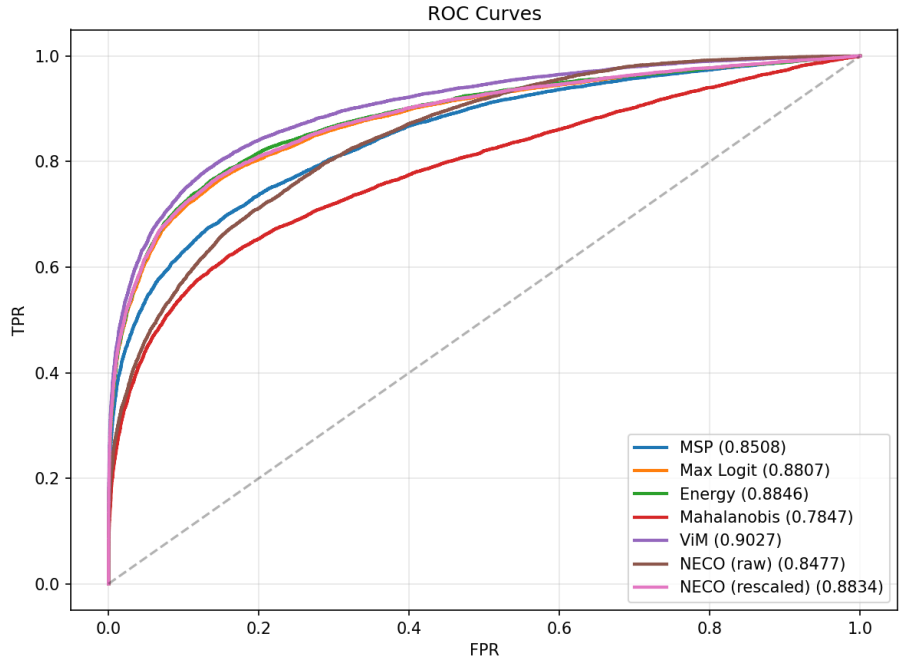


Figure 3: ROC curves for all OOD detection methods. ViM (purple) dominates, while Mahalanobis (red) clearly underperforms.

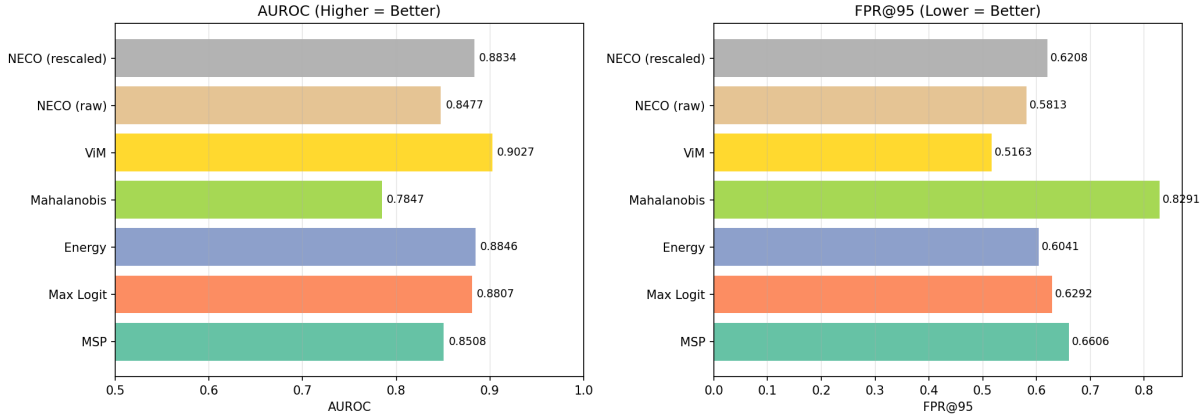


Figure 4: Summary bar chart: AUROC (left, higher is better) and FPR@95 (right, lower is better) for all methods.

4 Neural Collapse: NC1–NC4

Neural Collapse refers to a geometric structure that emerges in the last layer of a neural network once it has been trained long enough to reach near-zero training error. During this terminal phase, the features and classifier weights organize themselves into a highly symmetric configuration described by four properties.

We compute all Neural Collapse metrics on the training set, which is the correct setup: Neural Collapse describes how the network organizes the representations of data it has memorized, so we need to look at the training features.

4.1 NC1 — Variability Collapse

The first property says that features of the same class become tightly clustered around their class mean — the within-class variability essentially vanishes. We measure this with:

$$\text{NC1} = \frac{1}{C}\,\text{tr}\Big(\Sigma_W \Sigma_B^{\dagger}\Big)$$

where Σ_W is the average within-class covariance and Σ_B is the between-class covariance. A value of NC1 close to 0 means the features of each class have collapsed to a single point.

Results. We obtain NC1 = 0.509, which indicates significant collapse though not complete. The eigenvalue spectrum in Figure 5 (right panel) makes this visually clear: on a log scale, the eigenvalues of Σ_W (red) sit well below those of Σ_B (blue) across all components, confirming that within-class spread is much smaller than the separation between classes.

4.2 NC2 — Convergence to Simplex ETF

Once features have collapsed to their class means, NC2 tells us how those means are arranged in feature space. They should form a *Simplex Equiangular Tight Frame* (ETF): all class means equally spaced on a hypersphere, with every pair separated by the same angle. For C classes, the pairwise cosine similarities should all equal:

$$\cos(\theta_{ij}) = \frac{-1}{C-1}$$

For our 100 classes, the target is $-1/99 \approx -0.0101$.

Results. We measure a mean pairwise cosine similarity of $\mathbf{-0.0100}$, which is almost exactly the theoretical value of -0.0101. The standard deviation across pairs is 0.1005, and the normalized standard deviation of class mean norms is 0.0514, meaning the norms are fairly uniform. The heatmap in Figure 5 (left panel) shows this for the first 20 classes: the off-diagonal values are small and scattered around zero, consistent with the ETF geometry.

4.3 NC3 — Self-Duality

NC3 captures a striking fact: each classifier weight vector w_c ends up pointing in essentially the same direction as the corresponding centered class mean $\tilde{\mu}_c = \mu_c - \mu_G$:

$$\frac{w_c}{\|w_c\|} \approx \frac{\tilde{\mu}_c}{\|\tilde{\mu}_c\|}$$

This means the classifier hasn't learned anything beyond what the feature geometry already encodes — the weights and the means are dual to each other.

Results. The per-class cosine similarities between w_c and $\tilde{\mu}_c$ are shown in Figure 5 (center panel). The mean is $\mathbf{0.963}$ with a standard deviation of just 0.010, and even the worst class still achieves 0.925. This very strong alignment confirms NC3.

4.4 NC4 — Nearest Class Centre Simplification

The last property is a practical consequence of the first three. If features collapse to their means, the means form a symmetric frame, and the classifier aligns with those means, then the learned classifier should be equivalent to a simple Nearest Class Center (NCC) rule:

$$\arg\max_c \; w_c^{\top} f(x) \approx \arg\min_c \; \|f(x) - \mu_c\|^2$$

We could replace the entire last layer with a distance computation and get nearly the same predictions.

Results. On the test set, the trained classifier and the NCC rule agree on **95.8**% of predictions. Their individual accuracies are also very close: 77.1% for the classifier and 76.9% for NCC. This confirms that the network has effectively learned a nearest-mean classifier, which is a remarkably simple decision rule for a model with 11 million parameters.

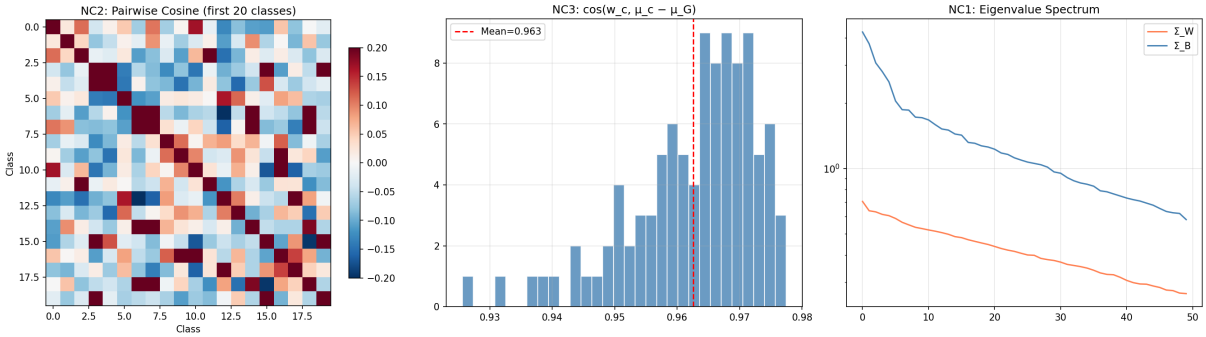


Figure 5: Neural Collapse analysis (NC1–NC4). **Left:** NC2 — pairwise cosine similarity between centered class means for the first 20 classes; values near zero match the Simplex ETF prediction. **Center:** NC3 — distribution of cosine similarities between classifier weights and class means (mean = 0.963). **Right:** NC1 — eigenvalue spectra of Σ_W (red) and Σ_B (blue) on a log scale; the consistent gap confirms variability collapse.

5 Neural Collapse: NC5 — ID/OOD Orthogonality

NC5 extends the Neural Collapse framework to out-of-distribution data. The intuition is natural: if the ID class means span a particular subspace (the ETF), then data that doesn't belong to any of those classes should be mapped *outside* that subspace — in fact, orthogonal to it. Formally:

$$\text{OrthoDev} = \frac{1}{C}\sum_{c=1}^{C}\left|\frac{\langle\tilde{\mu}_c,\,\tilde{\mu}_{\text{OOD}}\rangle}{\|\tilde{\mu}_c\|\cdot\|\tilde{\mu}_{\text{OOD}}\|}\right| \to 0$$

This property is important because it provides a theoretical justification for why feature-based OOD detection can work: OOD samples simply don't project onto the directions that matter for ID classification.

Results. Using SVHN as the OOD dataset, we measure an OrthoDev of **0.106** and a mean cosine (signed) of essentially zero (0.000077). Figure 6 (left) shows the per-class cosine similarities scattered around zero with no systematic bias, and Figure 6 (right) confirms that most absolute values are small, with the bulk of the distribution concentrated below 0.15. A few classes show slightly higher cosine values (up to ∼0.5), which is expected — perfect orthogonality is an idealization, and some SVHN features may accidentally align with certain CIFAR-100 class directions. Overall, the result clearly supports the NC5 hypothesis.

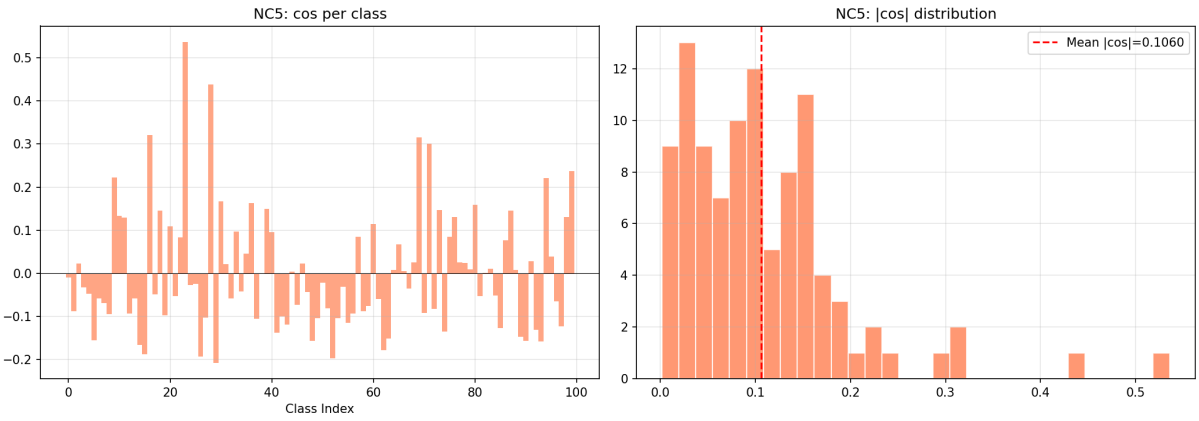


Figure 6: NC5 analysis. **Left:** cosine similarity between each centered ID class mean and the centered OOD mean — values are scattered around zero with no clear pattern. **Right:** distribution of $|\cos|$ values, concentrated near zero (mean = 0.106).

6 NECO — Neural Collapse Inspired OOD Detection

NECO directly exploits the NC5 property for OOD detection. Since OOD features are orthogonal to the ID subspace, we can quantify how much of a sample's features align with the principal subspace of ID training data.

6.1 Method

Let $h_\omega(x) \in \mathbb{R}^d$ denote the penultimate layer representation. We first centre the ID training features by the global mean and compute their top-k principal components via PCA, forming a projection matrix $P \in \mathbb{R}^{d \times k}$ (we use $k = C - 1 = 99$). The NECO score is the fraction of the feature norm that lies within this ID subspace:

$$\text{NECO}(x) = \frac{\|P^\top \tilde{h}_\omega(x)\|}{\|\tilde{h}_\omega(x)\|}$$

where $\tilde{h}_\omega(x) = h_\omega(x) - \mu_G$ is the centred feature. For ID samples whose features lie within the principal subspace, this ratio is close to 1. For OOD samples whose features are orthogonal to this subspace (NC5), the ratio is closer to 0.

6.2 Rescaled Variant

The raw NECO score captures geometric alignment but ignores class-level discriminative information. A rescaled variant multiplies by the maximum logit to combine both signals:

$$\text{NECO}_{\text{rescaled}}(x) = \text{NECO}(x) \times \max_c \; z_c(x)$$

This is analogous to how ViM combines feature-space geometry with logit-level confidence.

6.3 Results

The raw NECO score achieves an AUROC of **0.8477** and FPR@95 of **0.5813**. The score distribution (Figure 2, bottom-centre) shows that ID features have projection ratios concentrated near 0.97–0.99, while OOD features tend towards lower values around 0.93–0.96, though with significant overlap. Interestingly, while the AUROC is comparable to MSP, the FPR@95 of 0.5813 is notably better than MSP (0.6606) and even competitive with Energy (0.6041), suggesting that the geometric signal from NECO provides complementary information to logit-based methods.

The rescaled variant improves to an AUROC of **0.8834** and FPR@95 of **0.6208**, on par with Energy (0.8846). This confirms that combining the geometric insight of Neural Collapse (subspace projection ratio) with discriminative logit information yields a stronger signal. However, ViM still outperforms NECO-rescaled (0.9027 vs 0.8834), suggesting that the residual norm used by ViM captures complementary information beyond what the projection ratio provides.

It is worth noting that NECO was originally designed and evaluated on larger-scale settings with ViT and SwinV2 architectures on ImageNet, where Neural Collapse properties (especially NC5) tend to be stronger. On a smaller dataset like CIFAR-100 with a ResNet-18, the NC5 orthogonality is approximate (OrthoDev = 0.1060, not exactly 0), which limits NECO's separation power.

7 Conclusion

Training a ResNet-18 on CIFAR-100 to near-perfect training accuracy was enough to observe Neural Collapse in practice. The four classical properties all showed up: features of the same class clustered tightly around their mean (NC1 = 0.509), those means settled into a symmetric arrangement that almost exactly matches the theoretical Simplex ETF (mean pairwise cosine of -0.0100 vs. the predicted -0.0101), and the classifier weights ended up pointing in nearly the same directions as the class means (average cosine of 0.963). Also, with NC4, the neural collapse has been more confirmed. By replacing the learned classifier with a simple "pick the nearest class mean" rule gives the same prediction 95.8% of the time, with the same accuracy (77.1% vs. 76.9%). It is surprising that a model with 11 million parameters ultimately behaves like a nearest-mean classifier.

We also found support for NC5: features of OOD samples (from SVHN) turned out to be orthogonal to the ID class means, with an average absolute cosine of just 0.106. This is not perfect orthogonality, but it is close enough to be useful and it is exactly this property that makes methods like NECO possible.

For OOD detection, the hybrid method ViM performs best (AUROC 0.9027), followed by Energy (0.8846) and NECO-rescaled (0.8834). The progression from MSP to Energy to ViM illustrates how incorporating richer information — from softmax to logits to feature-space geometry — consistently improves detection. NECO provides a theoretically grounded alternative rooted in Neural Collapse, and notably achieves the second-best FPR@95 in its raw form (0.5813), demonstrating that the geometric signal from NC5 is a valuable complement to logit-based approaches.

The connection between Neural Collapse and OOD detection is the central insight of this work: the same geometric structure (ETF, orthogonality) that characterises ID representations also determines where OOD features land in feature space, enabling principled detection methods.